

Teaching Phonological Theory using Automatic Speech Recognition

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Roadmap

1. Motivation
2. Our approach
3. Procedure
4. Analysis
5. Conclusion

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Phonological theory

Teaching phonology and phonological theory is challenging

- Exemplary syllabus for introductory course:

Phonetic preliminaries
Phonemes
Features
Phonological alternations
Syllables
Stress and Intonation

Table 1: Basic schedule for phonology intro courses

- Contemporary phonological theories may be covered if time permits
- Emphasis on abstract sound patterns and problem solving
- How to best teach abstract concepts? e.g. programming? (Tucker et al., 2024)

Phonological theory

- Phonology requires several abstract theoretical concepts right from the start (phonemes, allophones, representations, rules...)
- If several phonological theories are covered: **confusing!**
- Approach involving aspects of sound structure and problem sets has not changed much in nearly 40 years, e.g. Gussenhoven & Jacobs (2017), Hayes (2009), Katamba (1989), and Zsiga (2024)
- This approach is timeless and does a very good job in introducing the field's central ideas!
- **This approach, however, does not address the students' interests!**

Students' perspective

Students often question the use of learning phonological theory

- Too abstract, thus too disconnected from real language
- Too many theoretical approaches to cover all
- Less engagement → loss of interest

Students' perspective

How to best teach abstract subjects such as phonology?

- Skill-based teaching (Zuraw et al., 2019)
- Project-based teaching (Kakolu Ramarao et al., 2024)
- **Increasing student engagement → better learning**

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Our approach I

- Our approach made use of phonetic data and Automatic Speech Recognition (ASR)
- Central idea: Have students record their **own non-native English speech** and **evaluate output of ASR** transcriptions using **phonological rules and features**
 - Involve students' English pronunciation
 - Address students' interest in language technology/computational linguistics
 - Connects with phonetic research on non-native language ASR (Emara & Shaker, 2024)
- We taught a 2 hour/week course for 13 weeks in summer 2025
- Elective course on Applied Phonology for German students
- Undergraduate students in General Linguistics + English & American Studies
- 20 registered students; 14 successfully finished the course

Our approach II

- Combined content from three fields
 1. Introduction to Acoustic Phonetics
 2. Introduction to Automatic Speech Recognition
 3. Introduction to Phonology

Central aspects

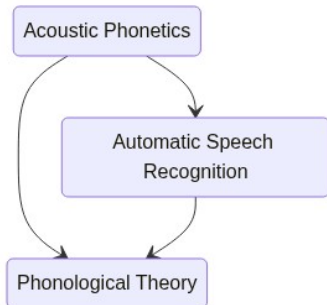


Figure 1: Combined approach

Our syllabus

#	Topic
01	Introduction & Logistics
-	HOLIDAY (no class)
02	Guided lab tour
03	Recordings I
04	Recordings II
05	Huggingface & Transcription
06	Introduction to Praat
-	STUDY WEEK (no class)
07	Phonology I: Recap
08	Phonology II: Processes
09	Phonology III: Features
10	Analysis I: Quantitative
11	Analysis II: Qualitative
12	Analysis III: Comparison
13	Wrap Up & Term paper

Table 2: Course outline

Material

- Sentences from **Artie bias corpus** (Meyer et al., 2020)
 - Natural sentences used for ASR benchmarking and detecting demographic bias
- Examples:
 - *No, we don't laminate our business cards.*
 - *Love is a fluid concept, and I'm not sure if androids get that.*
 - *The python wasn't a snake, it was a programming language.*
 - *Two-wave sources are perfectly coherent if they have a constant phase difference and the same frequency.*
- 70 sentences per speaker

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Procedure I

- Students were grouped based on their native language(s)
- Students recorded each other in our phonetics lab using *Audacity* (Audacity Team, 2025)
- We assigned ASR models to different students to compare model-performance on non-native speech
 - We selected models that differ in size, architecture, and how faithful the transcriptions were to the signal

Model	Labelled training data	Parameter size
Whisper-small (Radford et al., 2023)	680k hours	244 million
Whisper-large (Radford et al., 2023)	680k hours	1550 million
Crisper-Whisper (Wagner, Thallinger, & Zusag, 2024)	680k hours	finetuned whisper model
Parakeet (Sekoyan et al., 2025)	670k hours	600 million

Table 3: Overview ASR models

Procedure II

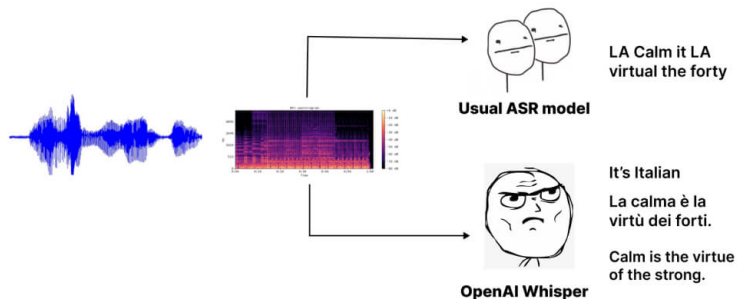


Figure 2: <https://www.superannotate.com/blog/openai-whisper-automatic-speech-recognition-system>

Procedure III

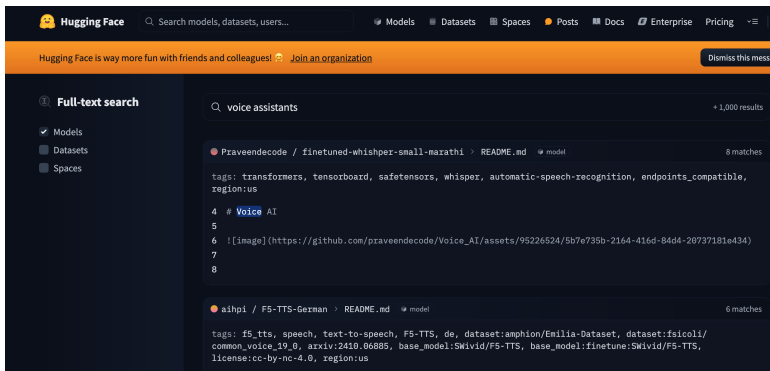


Figure 3: Testing Huggingface models

Procedure IV

- Students created annotated TextGrids based on ASR transcripts by using *Praat* (Boersma & Weenink, 2025)

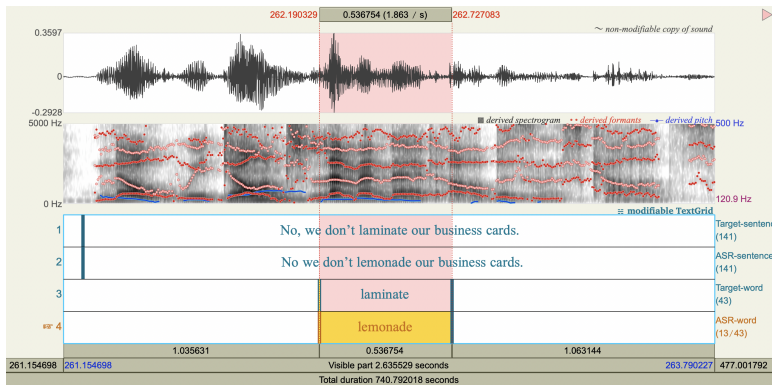


Figure 4: Acoustic analysis and transcription in Praat (Boersma & Weenink, 2025)

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Phonological analysis I

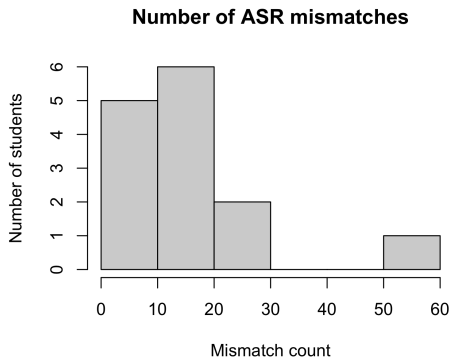


Figure 5: Number of word errors varied depending on ASR model and speaker data

- Annotations for 14 students
- Mean mismatches = 16.3 (SD $\pm$ 13.8)

Phonological analysis II

- Students with stronger English accent had more mismatches between target and ASR transcription
- Work around this by:
 - Using simpler models that produce more transcription errors
 - Using spontaneous speech instead of read speech
- Suprasegmentals and other phonological frameworks not covered

Phonological analysis III

- Students proposed phonological processes that they analyzed in analogy to rule-based phonological changes
- UR = Target pronunciation, SR = ASR transcription
- Phonological rules were written both as segments and features
- Rules were practiced with processes for Standard German using the *Pheatures* software (van Vugt, Hayes, & Zuraw, 2012)
- This facilitates exploration to see if differences in segmental inventory and processes will affect ASR transcriptions

Phonological analysis IV

Select Features

value

Feature

-

sonorant

value

Feature

+

voice

value

Transform Features

value

Feature

-

voice

value

Feature List

☐	Phoneme	syllabic ↓	stress	long	consonantal	sonorant	continuant	delayed release	approximant	tap	trill	nasal	voice	spread gl	const
☐	b → p	-	-	-	+	-	-	-	-	-	-	-	-	-	-
☐	d → t	-	-	-	+	-	-	-	-	-	-	-	-	-	-
☐	g → k	-	-	-	+	-	-	-	-	-	-	-	-	-	-
☐	v → f	-	-	-	+	-	+	+	-	-	-	-	-	-	-
☐	z → s	-	-	-	+	-	+	+	-	-	-	-	-	-	-
☐	ʁ → χ	-	-	-	+	-	+	+	-	-	-	-	-	-	-

Figure 6: Using *Pheatures* (van Vugt, Hayes, & Zuraw, 2012)

Online application (credits Taesung Hwang):

<https://pheatures.slam.phil.hhu.de/>

Phonological analysis V

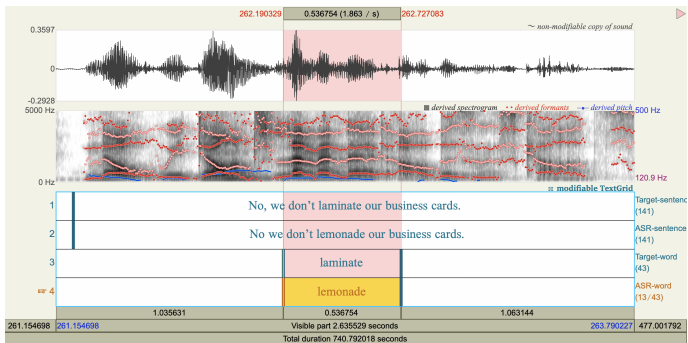


Figure 7: Acoustic analysis and transcription in Praat (Boersma & Weenink, 2025)

Target pronunciation	ASR transcription	Process (Vowel raising)
/æ/	[ɛ]	/æ/ → [ɛ]
[-back, +low, -high, -ATR]	[-back, -low, -high, -ATR]	[-back, +low, -high, -ATR] → [-low]

Table 4: Example phonological analysis

ASR Model comparison

For model comparison, quantitative analysis using standard ASR evaluation metrics were conducted

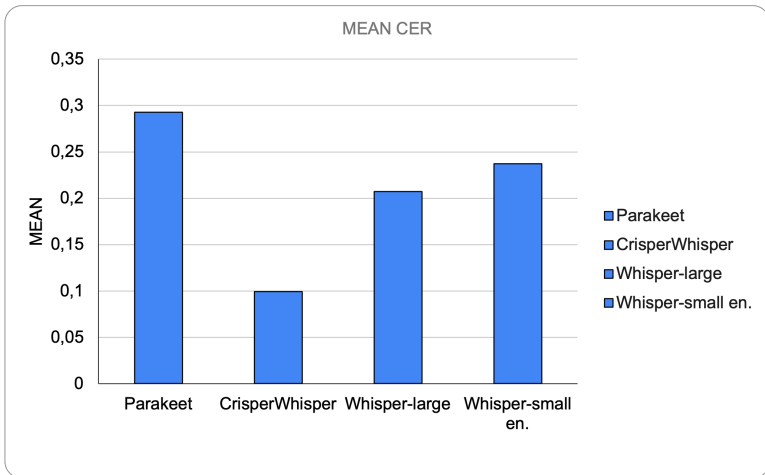


Figure 8: Results Character Error Rate (CER) by Model

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Learning outcomes I

Conceptual topics:

- Technical aspects of recording speech in a phonetics laboratory
- The basics of acoustic phonetics
- The basics of automatic speech recognition
- An introduction to the acoustic-phonetic software *Praat*
- Conceptual differences between phonetics and phonology in the context of German and English
- Analysis of phonological processes using phonological features
- Comparison of ASR model performance using standard ASR evaluation metrics

Learning outcomes II

Technical skills:

- Conducting speech recordings in a phonetics laboratory
- Using the acoustic-phonetic analysis software *Praat*
- Working with ASR models on the HuggingFace platform (Parakeet, Whisper-small, Whisper-large, CrisperWhisper)

Student feedback

Qualitative measure:

- Student find the hands-on aspects and course structure contributed to their learning success
- Several students found it hard to analyze ASR mismatches with rules and features

Quantitative measures:

Ihre Gesamtbewertung

All in all: How would you rate this event so far?

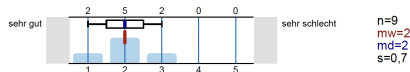


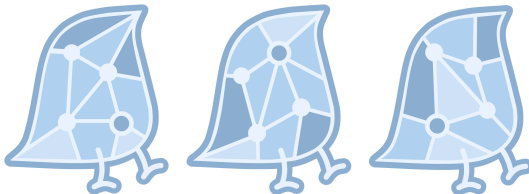
Figure 9: Overall student ratings (from 1-5, with 1 as best grade)

Takeaway

- We taught a two hour/week phonological theory class to undergraduate Linguistics and English students which made use of Automatic Speech Recognition (ASR)
- This project-based approach to learning about phonology held promising results
- We advocate for thinking about novel, timely ways to teach phonology
 - Incorporating language technology such as ASR + phonetic data may help connect students with phonology
 - Connecting theoretical notions with real data → more engagement and better learning

Acknowledgement

- Special thanks to our undergraduate students that participated in this project!
- Slamlab (<https://slam.phil.hhu.de/>) for providing the lab facility



End

Thank you for your attention!



- Scan QR code to access slides

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Appendix

Appendix

Results course evaluation I

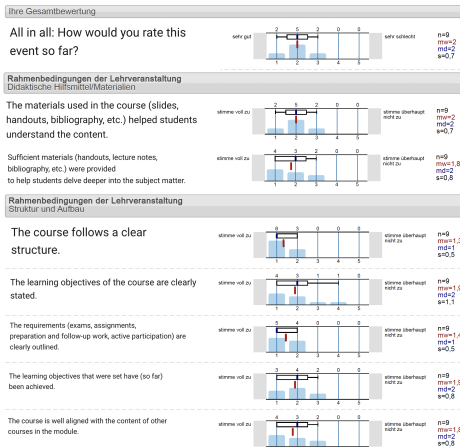


Figure 10: Course evaluation I (from 1-5, with 1 as best grade)

Results course evaluation II

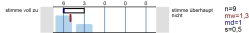
Lehrperson und Didaktik

Die/der Lehrende...

...makes the course interesting.



...always seems well-prepared.



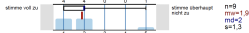
...expresses himself clearly and understandably.



...presented the individual course topics in such a way that I was able to understand them well.



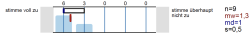
...often uses examples that help students understand the course material.



...provides constructive, individualized feedback (e.g., on presentations or questions).



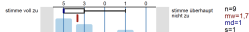
...responds appropriately to students' questions and suggestions.



...treats students in a friendly and respectful manner.



...encourages questions and active participation.



...has established a connection between the course content and the professional field or practical application.



...established a connection between the event's content and current research findings.



Figure 11: Course evaluation II (from 1-5, with 1 as best grade)

Results course evaluation III

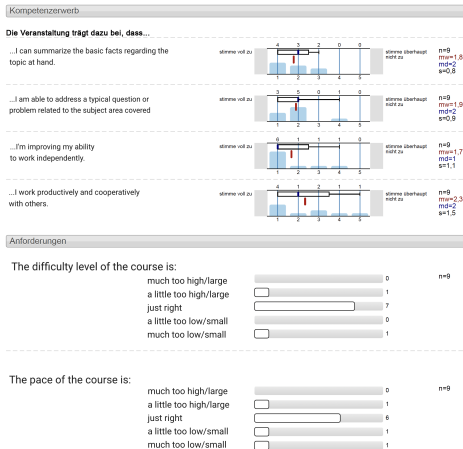


Figure 12: Course evaluation III (from 1-5, with 1 as best grade)

ASR metrics I

Character Error Rate (CER)

Calculates best alignment that minimizes the number of substitutions, insertions and deletions necessary to map the reference transcript to the hypothesis transcript.

$$CER = \frac{S + D + I}{N} \quad (1)$$

where,

S = number of substitutions

D = number of deletions

I = number of insertions

N = total number of characters in reference

ASR metrics II

Word Error Rate (CER)

Calculates best alignment that minimizes the number of substitutions, insertions and deletions necessary to map the reference transcript to the hypothesis transcript.

$$WER = \frac{S + D + I}{N} \quad (2)$$

where,

S = number of substitutions

D = number of deletions

I = number of insertions

N = total number of words in reference

Quantitative analysis ASR I

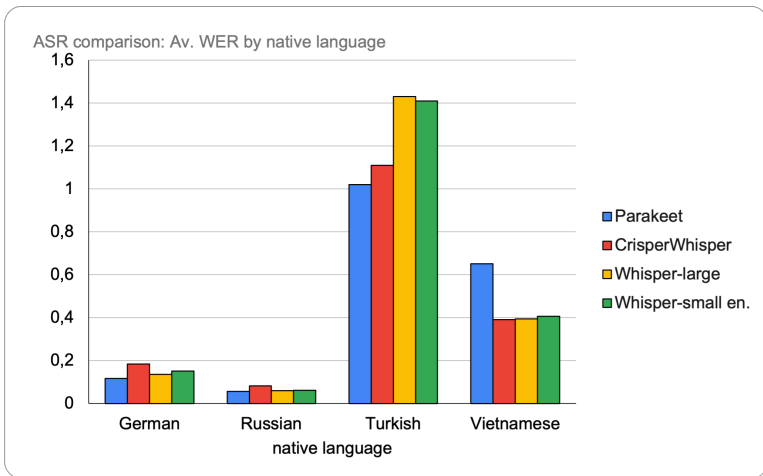


Figure 13: Results Word Error Rate (WER) – Language

Quantitative analysis ASR II

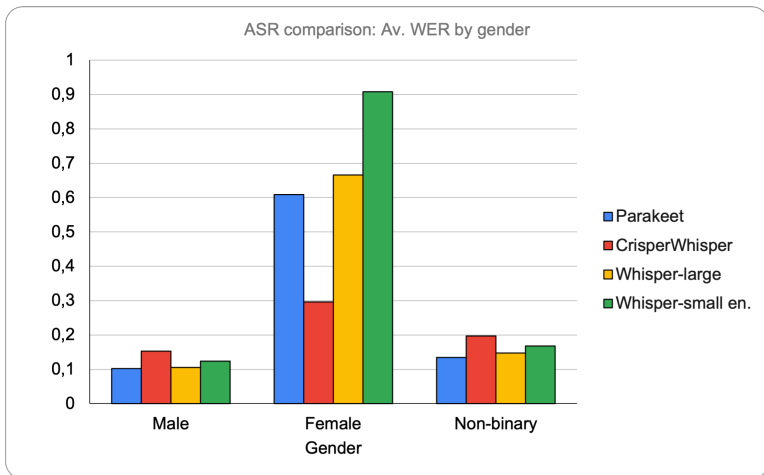


Figure 14: Results Word Error Rate (WER) – Gender

Quantitative analysis ASR III

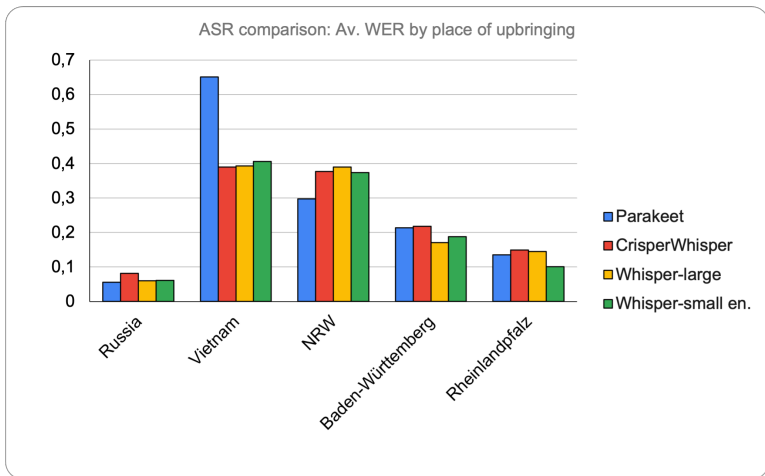


Figure 15: Results Word Error Rate (WER) – Upbringing